A Two-HCIZ Gaussian Matrix Model for Non-intersecting Brownian Bridges

Georgia Tech Stochastic Seminar

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A Gaussian Hermitian model with two
Harish–Chandra–Itzykson–Zuber insertions
reproducing the Karlin–McGregor law
for Dyson Brownian bridges



Alternative title

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(**9** people).



Brownian motion and Brownian bridge

Brownian motion (BM). Let $(B_t)_{t\geq 0}$ be standard Brownian motion on $\mathbb{R}$ with $B_0 = x$. Then for every $t > 0$,

$$B_t \,\big|\, (B_0 = x) \;\sim\; \mathcal{N}(x, t), \qquad \mathbb{P}(B_t \in \mathrm{d}z \mid B_0 = x) = p_t(x, z)\,\mathrm{d}z,$$

where the *heat kernel* (transition density) is

$$p_t(x, z) = \frac{1}{\sqrt{2\pi t}} \exp\Big(- \frac{(z-x)^2}{2t} \Big), \qquad t > 0.$$

Brownian bridge (BB) from x to y on $[0, 1]$ Karlin–McGregor (1959)

$$\mathbb{P}\big(X_t \in \mathrm{d}z \,\big|\, X_0 = x,\, X_1 = y\big) = \frac{p_t(x, z)\, p_{1-t}(z, y)}{p_1(x, y)}\,\mathrm{d}z.$$



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Brownian bridge (BB) as a Doob h-transform. Fix an endpoint y at time 1 and define, for $0 \leq s < 1$,

$$h(s, u) := p_{1-s}(u, y).$$

For $0 \leq s < t < 1$, the bridge transition density is the Doob transform

$$p^{\mathrm{BB}}_{s,t}(u, z) = p_{t-s}(u, z)\, \frac{h(t, z)}{h(s, u)} = p_{t-s}(u, z)\, \frac{p_{1-t}(z, y)}{p_{1-s}(u, y)}.$$

One-time marginal (bridge from x to y on $[0, 1]$). Taking $s = 0$ and $u = x$,

$$\mathbb{P}\big(B_t \in \mathrm{d}z \,\big|\, B_0 = x,\; B_1 = y\big) = p^{\mathrm{BB}}_{0,t}(x, z)\,\mathrm{d}z = \frac{p_t(x, z)\, p_{1-t}(z, y)}{p_1(x, y)}\,\mathrm{d}z.$$



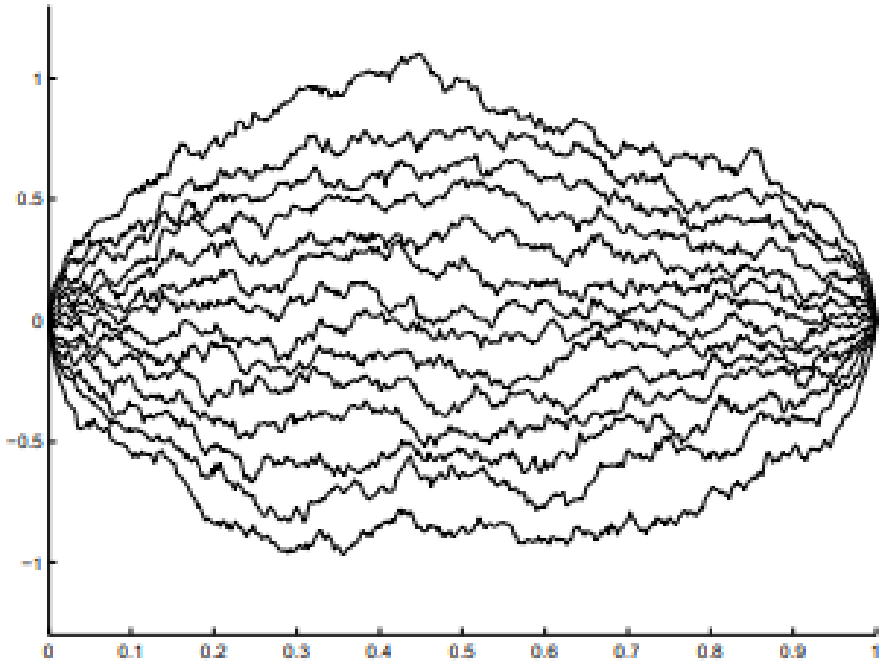


Figure 1: Non intersecting Brownian motions which start and end at 1 point. At any intermediate time the positions of the paths have the same distribution as the eigenvalues of a GUE matrix. The correlation kernel is built out of Hermite polynomials.

Karlin–McGregor for several starts/ends

Let starts be $a_1, \dots, a_n$ and ends be $b_1, \dots, b_n$ (distinct first; multiplicities via confluent limits).

Karlin–McGregor (1959):

$$\mathbb{P}\big(\lambda_1(t) \in \mathrm{d}x_1, \dots, \lambda_n(t) \in \mathrm{d}x_n\big) \propto \det\big[p_t(a_i, x_j)\big]_{i,j=1}^{n} \, \det\big[p_{1-t}(x_i, b_j)\big]_{i,j=1}^{n} \, \prod_{j=1}^{n} \mathrm{d}x_j,$$

restricted to a condition $x_1 < \cdots < x_n$. Explicitly,

$$\rho_{\mathrm{KM}}(\boldsymbol{x}) \propto \det\big[\mathrm{e}^{a_i x_j / t}\big] \, \det\big[\mathrm{e}^{b_i x_j / (1-t)}\big] \, \exp\Big(- \frac{1}{2\sigma^2} \sum_{j=1}^{n} x_j^2 \Big).$$



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RMT connection (Dyson, 1962)

For $x = y = 0$ the eignvelues of the Hermitian Gaussian ensemble has the same density, namely for $M \in \mathbb{H}(n)$

$$\mathrm{d}\mu(M) \propto \exp\Big(- \frac{1}{2\sigma^2} \operatorname{Tr}(M^2) \Big) \mathrm{d}M, \quad \sigma^2 = t(1-t).$$

External-field model (P.Zinn-Justin 1990; E.Brézin, S.Hikami 1998, Bleher–Kuijlaars 2000s)

Fix a diagonal "source" $A \in \mathbf{M}(n)$ with real eigenvalues. **External source ensemble:**

$$\mathrm{d}\mu_{\mathrm{ext}}(M) = \propto \ \exp\Big(-\frac{1}{2\sigma^2}\,\mathrm{Tr}\,M^2 + \frac{1}{t}\,\mathrm{Tr}(AM)\Big)\mathrm{d}M.$$

Not unitarily invariant (preferred eigenbasis from A).

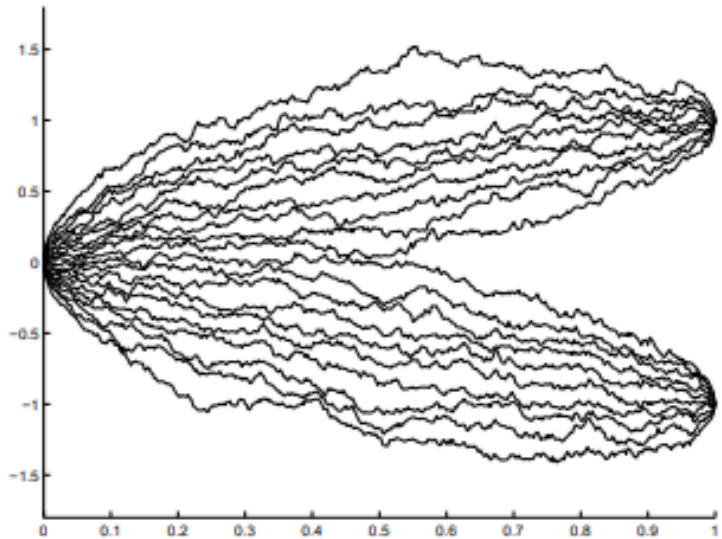


Figure 2: Non intersecting Brownian motions which start at 1 point and end at 2 different points. At any intermediate time the positions of the paths have the same distribution as the eigenvalues of a Gaussian unitary random matrix with external source. The correlation kernel is built out of multiple Hermite polynomials.

HCIZ integral (and repeated-eigenvalue / confluent case)

For Hermitian $X, Y \in \mathbb{H}(n)$ with eigenvalues $\boldsymbol{x}, \boldsymbol{y}$, the HCIZ formula (Harish–Chandra, 1957; Itzykson–Zuber, 1980) gives

$$\int_{\mathrm{U}(n)} \exp\bigl(\operatorname{Tr}(XUYU^{\dagger})\bigr)\,\mathrm{d}U = \Bigl(\prod_{k=1}^{n-1} k!\Bigr)\frac{\det\bigl[\mathrm{e}^{x_i y_j}\bigr]_{i,j=1}^{n}}{\Delta(\boldsymbol{x})\Delta(\boldsymbol{y})}.$$

Confluent case

If eigenvalues repeat, replace repeated rows/columns by derivatives:

$$\partial_x^{r-1}\mathrm{e}^{xy} = y^{r-1}\mathrm{e}^{xy},$$

and replace Vandermondes by confluent Vandermonde factors.

External-field eigenvalue density $\Rightarrow$ KM-type determinant

External source ensemble

$$\mathrm{d}\mu_{\mathrm{ext}}(M) = \propto \ \exp\Big(-\frac{1}{2\sigma^2}\,\mathrm{Tr}\,M^2 + \frac{1}{t}\,\mathrm{Tr}(AM)\Big)\mathrm{d}M.$$

Diagonalize $M = W\Lambda W^\dagger$ with $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$. Weyl integration gives

$$\mathrm{d}M \propto \Delta(\boldsymbol{\lambda})^2\,\mathrm{d}\boldsymbol{\lambda}\,\mathrm{d}W.$$

Integrate over W using HCIZ:

$$\int_{\mathrm{U}(n)} \exp\Big(\tfrac{1}{t}\,\mathrm{Tr}(AW\Lambda W^\dagger)\Big)\mathrm{d}W \propto \frac{\det[\mathrm{e}^{a_i\lambda_j/t}]}{\Delta(\boldsymbol{a})\Delta(\boldsymbol{\lambda})}.$$

Hence

$$\rho_{\mathrm{ext}}(\boldsymbol{\lambda}) \propto \Delta(\boldsymbol{\lambda})\ \det[\mathrm{e}^{a_i\lambda_j/t}]\ \exp\Big(-\frac{1}{2\sigma^2}\sum_{j=1}^{n}\lambda_j^2\Big).$$



The "unknown ensemble" in general (p, q)

Problem (E. Daems, A.B.J. Kuijlaars, 2005)

$$\rho_{\mathrm{KM}}(\boldsymbol{\lambda}) \propto \det\left[\mathrm{e}^{a_i \lambda_j / t}\right] \det\left[\mathrm{e}^{b_i \lambda_j / (1-t)}\right] \exp\Big(- \frac{1}{2\sigma^2} \sum \lambda_j^2 \Big).$$

What is the **parent matrix model** for $p, q \geq 2$?

Non-intersecting Brownian motions	Associated polynomials	Random matrix ensemble
1 starting point and 1 endpoint	Hermite polynomials	Gaussian unitary ensemble
1 starting point and $q \geq 2$ endpoints	multiple Hermite polynomials	Gaussian unitary ensemble with external source
$p \geq 2$ starting points and $q \geq 2$ endpoints	multiple Hermite polynomials of mixed type	unknown



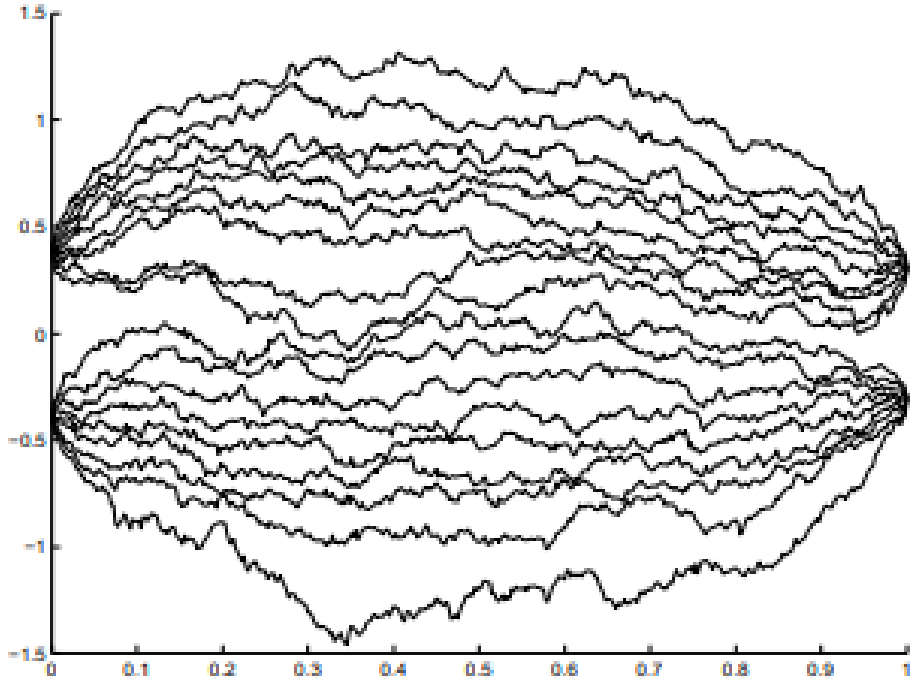


Figure 3: Non intersecting Brownian motions which start at 2 points and end at 2 points. At any intermediate time the positions of the paths are distributed according to a determinantal point process with a kernel that is built out of multiple Hermite polynomials of mixed type.

Idea

External source ensemble:

$$\mathrm{d}\mu_{\mathrm{ext}}(M) =\propto \ \exp\Big(-\frac{1}{2\sigma^2}\operatorname{Tr} M^2 + \frac{1}{t}\operatorname{Tr}(AM)\Big)\mathrm{d}M.$$

Diagonalize $M = W\Lambda W^\dagger$ with $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. Weyl integration gives

$$\mathrm{d}M \propto \Delta(\boldsymbol{\lambda})^2\, \mathrm{d}\boldsymbol{\lambda}\, \mathrm{d}W.$$

Integrate over W using HCIZ:

$$\int_{\mathrm{U}(n)} \exp\Big(\tfrac{1}{t}\operatorname{Tr}(AW\Lambda W^\dagger)\Big)\mathrm{d}W \propto \frac{\det[\mathrm{e}^{a_i\lambda_j/t}]}{\Delta(\boldsymbol{a})\Delta(\boldsymbol{\lambda})}.$$



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But

$$\int_{\mathrm{U}(n)} \mathrm{e}^{\frac{1}{t}\,\mathrm{Tr}(AUMU^\dagger)}\mathrm{d}U \propto \frac{\det[\mathrm{e}^{a_i\lambda_j/t}]}{\Delta(\boldsymbol{a})\Delta(\boldsymbol{\lambda})}$$



The two-HCIZ model: definition (measure + partition function)

Let $A, B \in \mathbb{H}(n)$ encode starts/ends and set $\sigma^2 = t(1-t)$. Define

$$\mathrm{d}\mu_{A,B,t}(M) = \frac{1}{Z_{A,B,t}} \mathrm{e}^{-\frac{1}{2\sigma^2}\operatorname{Tr} M^2} \Big(\int_{\mathrm{U}(n)} \mathrm{e}^{\frac{1}{t}\operatorname{Tr}(AUMU^\dagger)} \mathrm{d}U \Big) \Big(\int_{\mathrm{U}(n)} \mathrm{e}^{\frac{1}{1-t}\operatorname{Tr}(BVMV^\dagger)} \mathrm{d}V \Big) \mathrm{d}M,$$

with

$$Z_{A,B,t} := \int_{\mathbb{H}(n)} \mathrm{e}^{-\frac{1}{2\sigma^2}\operatorname{Tr} M^2} \Big(\int_{\mathrm{U}(n)} \mathrm{e}^{\frac{1}{t}\operatorname{Tr}(AUMU^\dagger)} \mathrm{d}U \Big) \Big(\int_{\mathrm{U}(n)} \mathrm{e}^{\frac{1}{1-t}\operatorname{Tr}(BVMV^\dagger)} \mathrm{d}V \Big) \mathrm{d}M.$$



Two-HCIZ $\Rightarrow$ KM

Diagonalize $M = W\Lambda W^\dagger$ and use Weyl:

$$\mathrm{d}M \propto \Delta(\boldsymbol{\lambda})^2\,\mathrm{d}\boldsymbol{\lambda}\,\mathrm{d}W.$$

Apply HCIZ twice:

$$\int_{\mathrm{U}(n)} \mathrm{e}^{\frac{1}{t}\,\mathrm{Tr}(AU\Lambda U^\dagger)}\mathrm{d}U \propto \frac{\det[\mathrm{e}^{a_i\lambda_j/t}]}{\Delta(\boldsymbol{a})\Delta(\boldsymbol{\lambda})}, \qquad \int_{\mathrm{U}(n)} \mathrm{e}^{\frac{1}{1-t}\,\mathrm{Tr}(BV\Lambda V^\dagger)}\mathrm{d}V \propto \frac{\det[\mathrm{e}^{b_i\lambda_j/(1-t)}]}{\Delta(\boldsymbol{b})\Delta(\boldsymbol{\lambda})}.$$

Cancel Vandermondes:

$$\Delta(\boldsymbol{\lambda})^2/(\Delta(\boldsymbol{\lambda})\Delta(\boldsymbol{\lambda})) = 1,$$

so

$$\rho(\boldsymbol{\lambda}) \propto \det[\mathrm{e}^{a_i\lambda_j/t}]\,\det[\mathrm{e}^{b_i\lambda_j/(1-t)}]\,\exp\Big(-\frac{1}{2\sigma^2}\sum\lambda_j^2\Big),$$

which matches KM formula.



If $A = aI$ and $B = bI \Rightarrow$ shifted Gaussian (GUE)

If $A = aI_n$ and $B = bI_n$, then both HCIZ factors collapse to scalar exponentials:

$$\int_{\mathrm{U}(n)} \mathrm{e}^{\frac{1}{t}\,\mathrm{Tr}(AUMU^\dagger)}\mathrm{d}U \propto \mathrm{e}^{\frac{a}{t}\,\mathrm{Tr}\,M}, \qquad \int_{\mathrm{U}(n)} \mathrm{e}^{\frac{1}{1-t}\,\mathrm{Tr}(BVMV^\dagger)}\mathrm{d}V \propto \mathrm{e}^{\frac{b}{1-t}\,\mathrm{Tr}\,M}.$$

Thus

$$\mathrm{d}\mu_{aI,bI,t}(M) \propto \exp\Big(-\frac{1}{2\sigma^2}\,\mathrm{Tr}\,M^2 + \Big(\frac{a}{t}+\frac{b}{1-t}\Big)\,\mathrm{Tr}\,M\Big)\mathrm{d}M.$$

Complete the square with

$$c = \sigma^2\Big(\frac{a}{t}+\frac{b}{1-t}\Big) = (1-t)a + tb, \qquad \widetilde{M} := M - cI_n,$$

so $\widetilde{M}$ is centered GUE with variance $\sigma^2 = t(1-t)$.



If $B = bI \Rightarrow$ external-field spectrum but unitary invariance

If $B = bI_n$, then $\operatorname{Tr}(BVMV^\dagger) = b \operatorname{Tr} M$ and the V-integral collapses:

$$\int_{\mathrm{U}(n)} \mathrm{e}^{\frac{1}{1-t}\operatorname{Tr}(BVMV^\dagger)} \mathrm{d}V \propto \mathrm{e}^{\frac{b}{1-t}\operatorname{Tr} M}.$$

Hence

$$\mathrm{d}\mu_{A,bI,t}(M) \propto \mathrm{e}^{-\frac{1}{2\sigma^2}\operatorname{Tr} M^2} \left(\int_{\mathrm{U}(n)} \mathrm{e}^{\frac{1}{t}\operatorname{Tr}(AUMU^\dagger)} \mathrm{d}U \right) \mathrm{e}^{\frac{b}{1-t}\operatorname{Tr} M}\, \mathrm{d}M,$$

so up to a shift

$$\mathrm{d}\mu_{A,bI,t}(\widetilde{M}) \propto \mathrm{e}^{-\frac{1}{2\sigma^2}\operatorname{Tr} \widetilde{M}^2} \left(\int_{\mathrm{U}(n)} \mathrm{e}^{\frac{1}{t}\operatorname{Tr}(AU\widetilde{M}U^\dagger)} \mathrm{d}U \right) \mathrm{d}\widetilde{M}.$$

Difference

Eigenvalue law matches the usual one-source external-field spectrum, while the **full matrix law remains unitarily invariant**.



Angular statistics: what changes and what does not

Spectral observables

Determined by $\rho(\boldsymbol{\lambda})$: density, correlation kernels, gap probabilities, ...

Angular observables

External-source ensembles select a preferred basis, while two-HCIZ is invariant: $M \mapsto WMW^\dagger \quad \Rightarrow \quad \mu_{A,B,t}$ unchanged. Thus, conditional on the spectrum, eigenvectors are Haar distributed.

$$M = U\Lambda U^\dagger, \qquad \Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n), \quad u_j := Ue_j.$$

Fix a unit vector $v \in \mathbb{C}^n$. Define the *local spectral measure*

$$\mu_v := \sum_{j=1}^{n} |\langle v, u_j \rangle|^2 \, \delta_{\lambda_j}.$$

Equivalently, for any bounded Borel test function f,

$$\langle v, f(M)v \rangle = \int f(\lambda) \, \mu_v(\mathrm{d}\lambda) = \sum_{j=1}^{n} f(\lambda_j) \, |\langle v, u_j \rangle|^2.$$

Local spectral measure

ETH / RMT interpretation

In ETH / quantum chaos, $\langle v, f(H)v\rangle$ is a canonical *local observable* (energy windowing via f). In RMT, it is the same object behind diagonal resolvent entries $G_{vv}(z) = \langle v, (M - zI)^{-1}v\rangle$.

What the two-HCIZ model gives (angular law)

Because $\mu_{A,B,t}$ is unitarily invariant, conditional on Λ the eigenbasis U is Haar on $U(n)$. Hence the weights $w_j := |\langle v, u_j\rangle|^2$ satisfy

$$(w_1, \dots, w_n) \,\Big|\, \Lambda \;\sim\; \mathrm{Dirichlet}(1, \dots, 1) \qquad \text{(complex case)},$$

so in particular

$$\mathbb{E}\big[\langle v, f(M)v\rangle \mid \Lambda\big] = \sum_{j=1}^{n} f(\lambda_j)\, \mathbb{E}[w_j \mid \Lambda] = \frac{1}{n}\sum_{j=1}^{n} f(\lambda_j) = \frac{1}{n}\,\mathrm{Tr}\, f(M).$$



One-HCIZ collapse of $Z_{A,B,t}$ and corollaries

$$Z_{A,B,t} := \int_{\mathbb{H}(n)} \mathrm{e}^{-\frac{1}{2\sigma^2}\,\mathrm{Tr}\,M^2} \Big(\int_{\mathrm{U}(n)} \mathrm{e}^{\frac{1}{t}\,\mathrm{Tr}(AUMU^\dagger)} \mathrm{d}U \Big) \Big(\int_{\mathrm{U}(n)} \mathrm{e}^{\frac{1}{1-t}\,\mathrm{Tr}(BVMV^\dagger)} \mathrm{d}V \Big) \mathrm{d}M.$$



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Collapse identity

$$Z_{A,B,t} = (2\pi\sigma^2)^{\frac{n^2}{2}} \exp\Big(\frac{1-t}{2t} \operatorname{Tr} A^2 + \frac{t}{2(1-t)} \operatorname{Tr} B^2 \Big) \int_{\mathrm{U}(n)} \exp\big(\operatorname{Tr}(AWBW^\dagger) \big) \mathrm{d}W.$$



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Time reversal: $Z_{A,B,t} = Z_{B,A,1-t}$.



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Time derivative: $\partial_t \log Z_{A,B,t} = -\frac{1}{2t^2} \operatorname{Tr} A^2 + \frac{1}{2(1-t)^2} \operatorname{Tr} B^2 + \frac{n^2}{2} \Big(\frac{1}{t} - \frac{1}{1-t} \Big).$



Proof of the collapse

Swap order and set

$$C(U,V) := \frac{1}{t}U^\dagger AU + \frac{1}{1-t}V^\dagger BV, \qquad \frac{1}{t}\operatorname{Tr}(AUMU^\dagger) + \frac{1}{1-t}\operatorname{Tr}(BVMV^\dagger) = \operatorname{Tr}(CM).$$

Gaussian identity:

$$\int_{\mathbb{H}(n)} \exp\Big(-\frac{1}{2\sigma^2}\operatorname{Tr} M^2 + \operatorname{Tr}(CM)\Big)\mathrm{d}M = (2\pi\sigma^2)^{\frac{n^2}{2}} \exp\Big(\frac{\sigma^2}{2}\operatorname{Tr} C^2\Big).$$

Expand:

$$\operatorname{Tr} C^2 = \frac{1}{t^2}\operatorname{Tr} A^2 + \frac{1}{(1-t)^2}\operatorname{Tr} B^2 + \frac{2}{t(1-t)}\operatorname{Tr}(AWBW^\dagger), \quad W := UV^\dagger,$$

and use Haar invariance to reduce $\iint F(UV^\dagger)\mathrm{d}U\mathrm{d}V = \int F(W)\mathrm{d}W$.



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and use Haar invariance to reduce $\iint F(UV^\dagger)\mathrm{d}U\mathrm{d}V = \int F(W)\mathrm{d}W$.

Corollary

$$\mathbb{E}[\operatorname{Tr} M] = (1-t)\operatorname{Tr} A + t\operatorname{Tr} B, \qquad \mathbb{E}[M] = \Big((1-t)\frac{\operatorname{Tr} A}{n} + t\frac{\operatorname{Tr} B}{n}\Big)I_n,$$

$$\mathbb{E}[\operatorname{Tr} M^2] = n^2\sigma^2 + (1-t)^2\operatorname{Tr} A^2 + t^2\operatorname{Tr} B^2 + 2t(1-t)\frac{(\operatorname{Tr} A)(\operatorname{Tr} B)}{n}.$$

Exact law of $\operatorname{Tr} M$

Let $M \sim \mu_{A,B,t}$ and $\sigma^2 = t(1-t)$. Then

$$\operatorname{Tr} M \sim \mathcal{N}\Big((1-t)\operatorname{Tr} A + t \operatorname{Tr} B,\ \sigma^2 n\Big).$$

Equivalently, for the non-intersecting Brownian bridges at time t,

$$\sum_{j=1}^{n} \lambda_j(t) \sim \mathcal{N}\Big((1-t)\operatorname{Tr} A + t \operatorname{Tr} B,\ t(1-t) n\Big).$$

In particular, for all $r > 0$,

$$\mathbb{P}\Big(|\operatorname{Tr} M - \mathbb{E}[\operatorname{Tr} M]| \geq r\Big) \leq 2\exp\Big(-\frac{r^2}{2\sigma^2 n}\Big).$$



Gaussian-mixture representation and concentration

Let $M \sim \mu_{A,B,t}$ and $\sigma^2 = t(1-t)$.

1. **(Exact Gaussian mixture; square-completion.)** Let $U, V \sim \mathrm{Haar}(U(n))$ be independent and let G be an independent centered Hermitian Gaussian with density proportional to $\exp(-\mathrm{Tr}(G^2)/(2\sigma^2))\,\mathrm{d}G$. Define

$$C(U,V) := \frac{1}{t} U^\dagger A U + \frac{1}{1-t} V^\dagger B V.$$

   Then the conditional law of M given (U,V) is affine Gaussian:

$$M \mid (U,V) \overset{d}{=} \sigma^2\, C(U,V) + G.$$

2. **(Sharp conditional subGaussian concentration.)** Let $F : \mathrm{H}(n) \to \mathbb{R}$ be L-Lipschitz w.r.t. $\|\cdot\|_{\mathrm{F}}$:

$$|F(X) - F(Y)| \le L \|X - Y\|_{\mathrm{F}} \qquad \forall\, X, Y \in \mathrm{H}(n).$$

   Then for every $r \ge 0$,

$$\mathbb{P}\Big(\big| F(M) - \mathbb{E}[F(M) \mid U,V] \big| \ge r \;\Big|\; U,V \Big) \le 2 \exp\Big(- \frac{r^2}{2\sigma^2 L^2} \Big).$$



Gaussian-mixture representation and concentration

Let $M \sim \mu_{A,B,t}$ and $\sigma^2 = t(1-t)$.

1. The conditional law of M given (U, V) is affine Gaussian:

$$M \mid (U, V) \overset{d}{=} \sigma^2\, C(U, V) + G.$$

2. **(Sharp conditional subGaussian concentration.)** Let $F : \mathrm{H}(n) \to \mathbb{R}$ be L-Lipschitz w.r.t. $\|\cdot\|_{\mathrm{F}}$:

$$|F(X) - F(Y)| \le L\|X - Y\|_{\mathrm{F}} \qquad \forall\, X, Y \in \mathrm{H}(n).$$

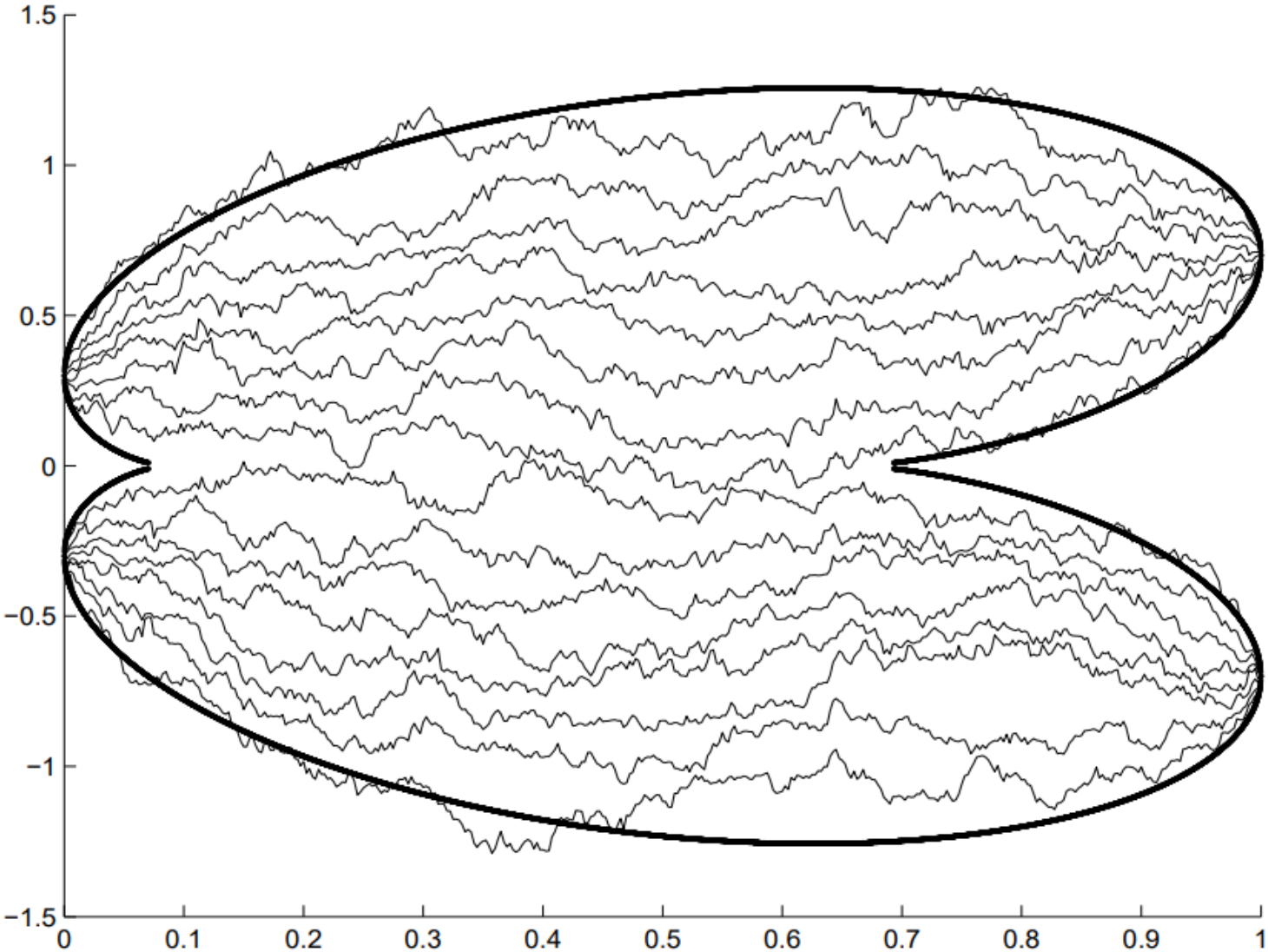
   Then for every $r \ge 0$,

$$\mathbb{P}\Big(\big|F(M) - \mathbb{E}[F(M) \mid U, V]\big| \ge r\Big) \le 2\exp\Big(-\frac{r^2}{2\sigma^2 L^2}\Big).$$

3. **(Spectral Lipschitz consequence.)** If $F(M) = \lambda_{\max}(M)$, then F is 1-Lipschitz w.r.t. $\|\cdot\|_{\mathrm{F}}$, hence the above holds with $L = 1$.



General (p, q)



One-time correlations (fixed t) & determinantal structure

Fix $t \in (0,1)$ and let $\boldsymbol{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ denote the positions of the n non-intersecting bridges at time t. Let $\rho_n(\boldsymbol{x})$ be the *one-time joint density*.

Definition (one-time k-point correlation function)

For $k = 1, \dots, n$, define

$$\rho^{(k)}(x_1, \dots, x_k) := \frac{n!}{(n-k)!} \int_{\mathbb{R}^{n-k}} \rho_n(x_1, \dots, x_n)\, \mathrm{d}x_{k+1} \cdots \mathrm{d}x_n,$$

interpreting ρ_n as 0 outside $x_1 < \cdots < x_n$.

Determinantal point process at fixed time

There exists a correlation kernel $K_t(x, y)$ such that for every k,

$$\rho^{(k)}(x_1, \dots, x_k) = \det\big[\, K_t(x_i, x_j) \,\big]_{i,j=1}^{k}.$$



Kernels and universality: bulk, edge, cusps, outliers

- **Bulk** $\Rightarrow$ **sine kernel** (Johansson, 2005; Deift–Gioev, 2009)

$$K_{\sin}(x,y) = \frac{\sin \pi(x-y)}{\pi(x-y)}.$$

- **Soft edge** $\Rightarrow$ **Airy kernel** (Tracy–Widom, 1994)

$$K_{Ai}(x,y) = \int_0^\infty Ai(x+s)Ai(y+s)\,\mathrm{d}s = \frac{Ai(x)Ai'(y) - Ai'(x)Ai(y)}{x-y}.$$

- **Cusp / merging cuts** $\Rightarrow$ **Pearcey kernel** (Brézin–Hikami, 1998)

$$K_{\mathrm{Pe}}(x,y) = \int_0^\infty p(x,u)\,q(y,u)\,\mathrm{d}u,$$

- **Outliers** $\Rightarrow$ **BBP transition kernel** (Baik–Ben Arous–Peche, 2005)

$$K_{BBP}^{(r)}(x,y) = K_{Ai}(x,y) + \sum_{a,b=1}^{r} \psi_a(x)\left[(\mathbf{I}-\mathbf{M})^{-1}\right]_{ab}\phi_b(y),$$

Haven't discussed

Mixed multiple orthogonal polynomials:

$$w_\ell^{(A)}(x) = \mathrm{e}^{a_\ell x/t}\,\mathrm{e}^{-x^2/(2\sigma^2)}, \qquad w_k^{(B)}(x) = \mathrm{e}^{b_k x/(1-t)}\,\mathrm{e}^{-x^2/(2\sigma^2)}.$$

Mixed-type MOP ensemble $\Rightarrow$ $(p+q)\times(p+q)$ RHP encodes the kernel and asymptotics (Kuijlaars, 2010; Deift–Zhou, 1993).

Ward identities (Schwinger, 1951; Dyson, 1962).

$$0 = \int_{\mathbb{H}(n)} \frac{\partial}{\partial M_{ij}} \Big(\Phi(M)\,\mathrm{e}^{-\frac{1}{2\sigma^2}\operatorname{Tr} M^2} \cdots \Big) \mathrm{d}M.$$

Choosing $\Phi(M) = \operatorname{Tr}\left((zI - M)^{-1}\right)$ gives resolvent loop equations and Virasoro-type constraints (Eynard, 2016).

2D Toda τ-function: $\mathrm{t}_m^{(+)} = \frac{1}{m}\operatorname{Tr} A^m$, $\mathrm{t}_m^{(-)} = \frac{1}{m}\operatorname{Tr} B^m$, then the HCIZ $\tau_n := \int_{\mathrm{U}(n)} \exp\left(\operatorname{Tr}(AWBW^\dagger)\right)\mathrm{d}W$ satisfies

$$\frac{\partial^2}{\partial t_1^{(+)}\partial t_1^{(-)}} \log \tau_n = \frac{\tau_{n+1}\tau_{n-1}}{\tau_n^2}.$$



Further directions

1. **Import the external-source toolbox.**
   - RHP steepest descent $\Rightarrow$ bulk/edge/cusp universality, phase diagrams, critical kernels,
   - loop equations / Ward identities $\Rightarrow$ constraints and PDEs for log-partition functions,
   - spiked/outlier regimes (BBP-type transitions) in appropriate scaling limits.
2. **Schur functions and Schur processes.** Use character expansions of HCIZ:

$$\int_{\mathrm{U}(n)(n)} e^{\mathrm{Tr}(AUBU^\dagger)}\,\mathrm{d}U \;=\; \sum_{\lambda} c_\lambda\, s_\lambda(A)\, s_\lambda(B),$$

   leading to *Schur-measure / Schur-process* structures (Okounkov, Reshetikhin, early 2000s), and to explicit formulas for Schur insertions / moments under the two-HCIZ measure.
3. **Enumerative geometry / Hurwitz counting.** The HCIZ integral (and its logarithm) admits expansions whose coefficients encode *double Hurwitz numbers* / branched cover counts (Hurwitz 1891; modern developments in the 2000s), connecting the two-HCIZ normalization and deformations to enumerative invariants and topological recursion.
4. **Beyond one-time: multi-time, Markov structure, and limits.**
5. **Angular observables** Since the two-HCIZ matrix law is unitarily invariant, conditional on Λ the eigenvectors are Haar; this makes Dirichlet calculus available for the spectral measures and related ETH diagnostics.



Thank you!

Questions / comments welcome.

